# Операционные системы

## Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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# 1. 1. Цели и задачи работы

## 1.1 Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

# 1.2 Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

## 2. 2. Процесс выполнения лабораторной работы

# 2.1 Перенаправление ввода-вывода

```
regenmammedov@regenmammedov:~$ ls /etc/ > file.txt
regenmammedov@regenmammedov:~$ ls >> file.txt
regenmammedov@regenmammedov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

**Рисунок 1:** Запись в файл

```
regenmammedov@regenmammedov:~$ grep .conf file.txt > conf.txt
regenmammedov@regenmammedov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

**Рисунок 2:** Поиск расширения .conf

# 2.3 Команда поиска

```
-personal/stage03/report/_resources/csl
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage03/report/bib/cite.bib
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/_resources/csl
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/bib/cite.bib
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/_resources/csl
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/bib/cite.bib
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/_resources/csl
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/bib/cite.bib
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/templat
e/config
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/templat
e/config/course
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/templat
e/config/course/computer-practice
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/templat
e/report/report/_resources/csl
/home/regenmammedov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/templat
e/report/report/bib/cite.bib
/home/regenmammedov/snap/hugo/common
/home/regenmammedov/snap/hugo/current
/home/regenmammedov/conf.txt
regenmammedov@regenmammedov:~$
```

**Рисунок 3:** Поиск файлов

# 2.4 Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

**Рисунок 4:** Поиск файлов

```
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$ find /etc -name "h*" -print | less
find: ‘/etc/audit’: Permission denied
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$ find ~ -name "log*" > logfile &
[1] 6551
regenmammedov@regenmammedov:~$
[1]+  Done                    find ~ -name "log*" > logfile
regenmammedov@regenmammedov:~$
^[[200~rm logfile
^[[201~regenmammedov@regenmammerm logfilelogfile
regenmammedov@regenmammedov:~$
```

**Рисунок 5:** Фоновый запуск процесса

```
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$ gedit &
[1] 6595
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$ ps | grep gedit
   6595 pts/1    00:00:00 gedit
regenmammedov@regenmammedov:~$ kill 6595
[1]+  Terminated              gedit
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$
```

**Рисунок 6:** Фоновый запуск и завершение процесса

# 2.7 Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents the GNU version of df.  df displays the amount of space
       available on the file system containing each file name argument.  If no file name  is
       given,  the space available on all currently mounted file systems is shown.  Space is
       shown in 1K blocks by default, unless the  environment  variable  POSIXLY_CORRECT  is
       set, in which case 512-byte blocks are used.

       If  an  argument is the absolute file name of a device node containing a mounted file
       system, df shows the space available on that file system rather than on the file sys-
       tem containing the device node.  This version of df cannot show the  space  available
       on  unmounted  file  systems, because on most kinds of systems doing so requires non-
       portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all  file  sys-
       tems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
 Manual page df(1) line 1 (press h for help or q to quit)
```

**Рисунок 7:** Справка по команде df

# 2.8 Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes rather than device usage; although the apparent size is
              usually smaller, it may be larger due to holes in ('sparse')  files,  internal
              fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
              1,048,576 bytes; see SIZE format below
 Manual page du(1) line 1 (press h for help or q to quit)
```

**Рисунок 8:** Запуск команды df

# 2.9 Команды df и du

```
regenmammedov@regenmammedov:~$
regenmammedov@regenmammedov:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89006452  65542028  58% /
devtmpfs         4011288        0   4011288   0% /dev
tmpfs            4035660       92   4035568   1% /dev/shm
tmpfs            1614268     1948   1612320   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035664     1540   4034124   1% /tmp
/dev/nvme0n1p3 155186176 89006452  65542028  58% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      176    806956   1% /run/user/1011
/dev/sr0         2677920  2677920         0 100% /run/media/regenmammedov/Fedora-WS-Live-43
regenmammedov@regenmammedov:~$
```

**Рисунок 9:** Справка по команде du

# 2.10 Команды df и du

```
12      ./snap/hugo/25676/.config/go/telemetry
12      ./snap/hugo/25676/.config/go
12      ./snap/hugo/25676/.config
12      ./snap/hugo/25676
0       ./snap/hugo/common
16      ./snap/hugo
16      ./snap
628     ./.texlive2025/texmf-var/luatex-cache/generic/names
68964   ./.texlive2025/texmf-var/luatex-cache/generic/fonts/otl
68964   ./.texlive2025/texmf-var/luatex-cache/generic/fonts
69592   ./.texlive2025/texmf-var/luatex-cache/generic
69592   ./.texlive2025/texmf-var/luatex-cache
69592   ./.texlive2025/texmf-var
69592   ./.texlive2025
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
1090744 .
regenmammedov@regenmammedov:~$
```

**Рисунок 10:** Запуск команды du

# 2.11 Команда поиска

```
e/report/scripts
/home/regenmammedov/snap
/home/regenmammedov/snap/hugo
/home/regenmammedov/snap/hugo/25676
/home/regenmammedov/snap/hugo/25676/.config
/home/regenmammedov/snap/hugo/25676/.config/go
/home/regenmammedov/snap/hugo/25676/.config/go/telemetry
/home/regenmammedov/snap/hugo/25676/.config/go/telemetry/local
/home/regenmammedov/snap/hugo/25676/.config/go/telemetry/upload
/home/regenmammedov/snap/hugo/common
/home/regenmammedov/.texlive2025
/home/regenmammedov/.texlive2025/texmf-var
/home/regenmammedov/.texlive2025/texmf-var/luatex-cache
/home/regenmammedov/.texlive2025/texmf-var/luatex-cache/generic
/home/regenmammedov/.texlive2025/texmf-var/luatex-cache/generic/names
/home/regenmammedov/.texlive2025/texmf-var/luatex-cache/generic/fonts
/home/regenmammedov/.texlive2025/texmf-var/luatex-cache/generic/fonts/otl
/home/regenmammedov/monthly
/home/regenmammedov/reports
/home/regenmammedov/reports/monthly
/home/regenmammedov/reports/monthly/monthly
/home/regenmammedov/ski.plases
/home/regenmammedov/ski.plases/equipment
/home/regenmammedov/ski.plases/plans
/home/regenmammedov/australia
/home/regenmammedov/play
/home/regenmammedov/play/games
/home/regenmammedov/play/games/play
regenmammedov@regenmammedov:~$
```

**Рисунок 11:** Поиск директорий

# 3. 3. Выводы по проделанной работе

## 3.1 Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.